

Higher-Order Asymptotics of the Discrepancy Function under Fixed Model and Distributional Misspecification

June 20, 2026

Purpose

The companion note `rmsea_asymptotics.tex` fixes the meaning of RMSEA by a Pitman drift: the misspecification is sent to zero at the $\sqrt{n}$ rate so the test statistic converges to a (noncentral) χ^2 and the noise inflation is the first-order Satorra–Bentler trace $\text{tr}(UT)$. This note drops the drift. It expands the fitted discrepancy $\hat{F}$ to order n^{-1} under *fixed* misspecification in both the covariance structure (σ_0 off the model) and the distribution ($\Gamma \neq$ normal theory), for a general minimum-discrepancy estimator. The point is to see what the local sequence silently discards and to ask whether GLS-RMSEA differs from normal-theory RMSEA at all once the drift is removed.

The single result is the bias expansion

$$\mathbb{E}[\hat{F}] = F_0 + \frac{1}{n} \left[\underbrace{\text{tr}(\tilde{U}\Gamma)}_{\text{generalized } df} + \underbrace{\beta_W}_{\text{weight influence, } \propto e} \right] + o(n^{-1}), \quad (1)$$

with two objects that the drift kills. The trace carries the *curvature-corrected* residual-weight matrix $\tilde{U} = V - V\dot{\sigma}B^{-1}\dot{\sigma}'V$, built from the population *Hessian* $B = A - K$ of the discrepancy rather than its Gauss–Newton part $A = \dot{\sigma}'V\dot{\sigma}$; the two agree only when the residual–curvature coupling K vanishes (correct specification, or a model linear in θ). The remainder β_W is the influence of an estimated weight; it is identically zero for a fixed weight (ULS) and vanishes as the residual $e \rightarrow 0$ for any estimator. Three corollaries read off (1): a curvature-corrected, robust bias correction for RMSEA

(Corollary 1); an honest Wald confidence interval that needs no noncentral- χ^2 inversion (Corollary 2); and the ML/GLS/ULS comparison (Section 6).

Two structural remarks orient the rest. First, $\tilde{U}$ uses the Hessian B where the first-order projector U uses A ; this is the fit-index sibling of the observed-Hessian bread that replaces Fisher information in the misspecification-robust standard errors of `misspec_observed_bread.tex`, and β_W is the fit-index sibling of the data-dependent-weight meat term there. Both corrections are zero at H_0 and leading-order off it; the order-promotion is the same theorem twice. Second, in the language of `aic_tic_rmsea_bias.tex`, $\text{tr}(\tilde{U}\Gamma)$ is to $\text{tr}(U\Gamma)$ as TIC's $\text{tr}(I^{-1}J)$ is to AIC's k , and then one order deeper: the curvature term is a correction past even the first-order robust (TIC-like) trace.

1 Setup

Stack the $p^* = p(p+1)/2$ distinct sample covariances as $s = \text{vech } \hat{\Sigma}$ (append means for a mean structure; the algebra is unchanged), with population limit $s \xrightarrow{p} \sigma_0$ and

$$z := \sqrt{n}(s - \sigma_0) \xrightarrow{d} N(0, \Gamma), \quad (2)$$

Γ the asymptotic covariance of the moments, normal-theory or not. The model is a smooth map $\theta \mapsto \sigma(\theta)$, $\theta \in \mathbb{R}^t$, $df = p^* - t$. A minimum-discrepancy estimator minimizes a quadratic form in the residual $r(\theta) = s - \sigma(\theta)$,

$$\hat{F} = \min_{\theta} r(\theta)' \hat{V} r(\theta), \quad \hat{V} \xrightarrow{p} V, \quad (3)$$

with weight $\hat{V}$ either fixed (ULS) or a smooth function of the data (GLS). Maximum likelihood is not exactly of this form; it is treated as the reweighted limit in Section 6.

Pseudo-true value and population residual. Let

$$\theta_* = \arg \min_{\theta} (\sigma_0 - \sigma(\theta))' V (\sigma_0 - \sigma(\theta)), \quad e := \sigma_0 - \sigma(\theta_*), \quad F_0 = e' V e. \quad (4)$$

Write $\dot{\sigma} = \partial\sigma/\partial\theta'$ at θ_* ($p^* \times t$, full column rank) and the model Hessian tensor $\ddot{\sigma}$, $(\ddot{\sigma})_{a,ij} = \partial^2\sigma_a/\partial\theta_i\partial\theta_j$. The first-order condition for (4) is the V -orthogonality of the residual to the tangent space,

$$\dot{\sigma}' V e = 0. \quad (5)$$

Under correct specification $e = 0$; under fixed misspecification $e = O(1)$ and (5) is all that survives of it.

Curvature objects. Define the Gauss–Newton matrix, the residual–curvature coupling, and the population Hessian of the discrepancy,

$$A := \dot{\sigma}' V \dot{\sigma}, \quad K_{ij} := (Ve)'(\ddot{\sigma})_{:,ij} = \sum_a (Ve)_a(\ddot{\sigma})_{a,ij}, \quad B := A - K = \frac{1}{2} \partial_{\theta\theta}^2 (e'Ve). \quad (6)$$

K is the only object in (6) that requires both a nonzero residual (Ve) and a curved model $(\ddot{\sigma} \neq 0)$; it is the misspecification signature. With B in hand define the standard residual projector and its curvature-corrected analogue

$$U := V - V \dot{\sigma} A^{-1} \dot{\sigma}' V, \quad \tilde{U} := V - V \dot{\sigma} B^{-1} \dot{\sigma}' V = V - V P V, \quad P := \dot{\sigma} B^{-1} \dot{\sigma}'. \quad (7)$$

U is a genuine residual projector ($U \dot{\sigma} = 0$, idempotent in the V -metric); $\tilde{U}$ is a perturbation of it that no longer annihilates the tangent ($\tilde{U} \dot{\sigma} = -V \dot{\sigma} B^{-1} K$), reducing to U as $K \rightarrow 0$. U is the object of every limit law in the drift regime (`rmsea_asymptotics.tex`); $\tilde{U}$ is its fixed-misspecification replacement.

Weight fluctuation. For an estimated weight expand

$$\hat{V} = V + n^{-1/2} \dot{V} + \frac{1}{2} n^{-1} \ddot{V} + o_p(n^{-1}), \quad (8)$$

where $\dot{V} = O_p(1)$ is linear in z (the derivative of the weight map along the moment fluctuation) and $\ddot{V} = O_p(1)$ is the second-order term. Write the residual-projected weight fluctuation

$$w := \dot{V} e, \quad (9)$$

an $O_p(1)$ vector that is identically zero for a fixed weight and is $O(\|e\|)$ otherwise. For GLS, $\dot{V}$ is driven by the raw moment fluctuation z ; for ML, by the projected fluctuation $\dot{\sigma} (\hat{\theta} - \theta_*)$ (Section 6).

2 The stochastic expansion

Theorem 1 (Second-order expansion of $\hat{F}$). *Assume (2), identification ($\dot{\sigma}$ full rank, B nonsingular), a weight that does not depend on θ (ULS, GLS, or any fixed/data weight), and the usual smoothness. Then*

$$\hat{F} = F_0 + n^{-1/2} (2e'Vz + e'\dot{V}e) + n^{-1} \Phi + o_p(n^{-1}), \quad (10)$$

with

$$\Phi = z' \tilde{U} z + 2w'(I - PV)z - w'Pw + \frac{1}{2}e'\ddot{V}e. \quad (11)$$

Proof. Write $u = \hat{\theta} - \theta_* = O_p(n^{-1/2})$ and split it as $u = n^{-1/2}\xi + O_p(n^{-1})$. Expand the residual at the estimator,

$$r(\hat{\theta}) = e + \rho_1 + \rho_2 + o_p(n^{-1}), \quad \rho_1 = n^{-1/2}z - \dot{\sigma}u, \quad \rho_2 = -\frac{1}{2}(\ddot{\sigma})[u, u],$$

where $(\ddot{\sigma})[u, u]$ has a -component $u'(\ddot{\sigma})_{:,a}u$. The estimating equation $\dot{\sigma}(\hat{\theta})'\hat{V}r(\hat{\theta}) = 0$, expanded to $O_p(n^{-1/2})$ with $\dot{\sigma}(\hat{\theta}) = \dot{\sigma} + (\ddot{\sigma})[u, \cdot]$ and (8), gives after using (5)

$$-Au + (\ddot{\sigma})[u, \cdot]'Ve + n^{-1/2}\dot{\sigma}'Vz + n^{-1/2}\dot{\sigma}'\dot{V}e = o_p(n^{-1/2}).$$

The second term is Ku by (6); collecting,

$$(A-K)u = n^{-1/2}\dot{\sigma}'(Vz + \dot{V}e) + o_p(n^{-1/2}) \implies \xi = B^{-1}\dot{\sigma}'(Vz + w), \quad u = n^{-1/2}\xi. \quad (12)$$

The weight fluctuation enters ξ only through $w = \dot{V}e$, hence only under misspecification. Now substitute into $\hat{F} = (e + \rho)'\hat{V}(e + \rho)$, $\rho = \rho_1 + \rho_2$, and use (5) twice. The cross term with ρ_1 collapses to the linear part, $2e'V\rho_1 = 2n^{-1/2}e'Vz$ (the $\dot{\sigma}u_1$ piece dies by (5)); the cross term with ρ_2 kills the second-order parameter increment and leaves only curvature, $2e'V\rho_2 = -u'Ku$; and $e'\dot{V}e$ contributes the weight terms $n^{-1/2}e'\dot{V}e + \frac{1}{2}n^{-1}e'\ddot{V}e$. Together with $\rho_1'V\rho_1 = n^{-1}(z'Vz - 2\xi'\dot{\sigma}'Vz + \xi' A\xi)$ and the $\dot{V}\rho_1$ cross term $2n^{-1}w'(z - \dot{\sigma}\xi)$, the $O(n^{-1})$ part is

$$\Phi = z'Vz - 2\xi'\dot{\sigma}'Vz + \xi'B\xi + 2w'(z - \dot{\sigma}\xi) + \frac{1}{2}e'\ddot{V}e,$$

where $\xi' A\xi - u'Ku/n^{-1}$ combined to $\xi'B\xi$. Insert $\xi = B^{-1}\dot{\sigma}'(Vz + w)$. With $P = \dot{\sigma}B^{-1}\dot{\sigma}'$ one gets $\xi'B\xi = (Vz + w)'P(Vz + w)$, $\xi'\dot{\sigma}'Vz = (Vz + w)'PVz$ and $\dot{\sigma}\xi = P(Vz + w)$. Expanding and cancelling,

$$\Phi = z'(V - VPV)z + 2w'(I - PV)z - w'Pw + \frac{1}{2}e'\ddot{V}e,$$

which is (11) with $\tilde{U} = V - VPV$. $\square$

Remark (The cancellation is the content). Everything hinges on the population FOC (5). It removes the tangent-direction sampling noise from the linear term (leaving $2e'Vz = 2e'Uz$, since $e'V = e'U$ by (5)) and removes the second-order parameter increment u_2 from the bias, so the bias needs ξ only to first order. What remains is genuinely a residual-space object, projected by $\tilde{U}$.

3 Bias and the curvature-corrected df

Corollary 1 (Generalized degrees of freedom). *Under the conditions of Theorem 1,*

$$\mathbb{E}[\hat{F}] = F_0 + \frac{1}{n} [\text{tr}(\tilde{U}\Gamma) + \beta_W] + o(n^{-1}), \quad \beta_W = 2\mathbb{E}[w'(I - PV)z] - \mathbb{E}[w'Pw] + \frac{1}{2}e'\mathbb{E}[\ddot{V}]e. \quad (13)$$

For a fixed weight $\beta_W \equiv 0$; for any estimator $\beta_W = O(\|e\|)$, so it vanishes at H_0 . Under correct specification $e = 0$, $K = 0$, $\tilde{U} = U$ and (13) reduces to $\mathbb{E}[\hat{F}] = \text{tr}(U\Gamma)/n + o(n^{-1})$, the Satorra–Bentler mean cdf = $\text{tr}(U\Gamma)$ of `rmsea_asymptotics.tex`, which is df when $V = \Gamma^{-1}$.

Proof. Take expectations in (10). The $n^{-1/2}$ term has mean zero ($\mathbb{E}z = 0$, $\mathbb{E}\dot{V} = 0$). For the n^{-1} term, $\mathbb{E}[z'\tilde{U}z] = \text{tr}(\tilde{U}\Gamma)$ by the quadratic-form mean and the remaining terms define β_W . $\square$

The estimand correction therefore generalizes the chain

$$df \longrightarrow \text{tr}(U\Gamma) \longrightarrow \text{tr}(\tilde{U}\Gamma) + \beta_W,$$

in increasing order of fidelity: df assumes a correctly weighted, correctly distributed null; $\text{tr}(U\Gamma)$ allows distributional misspecification but freezes the model at its tangent; $\text{tr}(\tilde{U}\Gamma) + \beta_W$ allows fixed structural misspecification, correcting the projector for curvature and adding the weight influence. A curvature-and-weight corrected RMSEA is

$$\hat{\varepsilon}_{\text{ho}} = \sqrt{\max\left(\frac{\hat{F}}{df} - \frac{\text{tr}(\hat{\tilde{U}}\hat{\Gamma}) + \hat{\beta}_W}{n df}, 0\right)}. \quad (14)$$

Remark (Curvature is real for SEM). $K \neq 0$ requires $\ddot{\sigma} \neq 0$. Factor and path models are nonlinear in θ (loadings multiply, paths compose), so $\ddot{\sigma} \neq 0$ generically and the curvature term is present whenever the model is wrong. It vanishes for models linear in θ (e.g. a free covariance matrix with linear equality constraints), where $\tilde{U} = U$ exactly even under misspecification.

4 The honest interval

Corollary 2 (CLT and Wald interval, no noncentral inversion). *If $F_0 > 0$, then from (10)*

$$\sqrt{n}(\hat{F} - F_0) \xrightarrow{d} N(0, \omega^2), \quad \omega^2 = \text{Var}(2e'Vz + e'\dot{V}e) = (2Ve + v_e)'\Gamma(2Ve + v_e), \quad (15)$$

where v_e is the vector representing the linear functional $z \mapsto e'\dot{V}e = v_e'z$; for a fixed weight $v_e = 0$ and $\omega^2 = 4e'V\Gamma V e = 4e'U\Gamma U e$. By the delta method, with $\varepsilon = \sqrt{F_0/df}$,

$$\sqrt{n}(\hat{\varepsilon} - \varepsilon) \xrightarrow{d} N\left(0, \frac{\omega^2}{4df F_0}\right), \quad (16)$$

giving a Wald interval for ε directly on the discrepancy scale.

Equation (16) is the fixed-misspecification interval: under a genuinely wrong model the statistic is asymptotically normal, not noncentral χ^2 , and the interval is a plain sandwich, not a CDF inversion. It complements rather than replaces the noncentral interval of `rmsea_asymptotics.tex`: that one is built for the near-null regime ($F_0 \downarrow 0$) where (16) degenerates (ω^2/F_0 blows up), while (16) is built for clearly-misspecified models where the noncentral inversion is least trustworthy. The crossover is exactly where neither device is comfortable, which is the honest statement of where RMSEA interval coverage is hard.

5 What the drift discards

Put back the Pitman sequence $e_n = n^{-1/2}\delta$ with δ a fixed off-model direction, the regime of `rmsea_asymptotics.tex`. Then $Ve_n = O(n^{-1/2})$ so $K = O(n^{-1/2}) \rightarrow 0$, hence $B \rightarrow A$ and $\tilde{U} \rightarrow U$; the weight influence $w = \dot{V}e_n = O_p(n^{-1/2}) \rightarrow 0$, so $\beta_W \rightarrow 0$; and the normal CLT (15) degenerates while $n\hat{F}$ acquires the (noncentral) χ^2 limit. The drift annihilates precisely the two corrections this note is about.

	fixed misspecification ($e = O(1)$)	Pitman drift ($e = n^{-1/2}\delta$)
limit law of $\hat{F}$	normal, $\sqrt{n}(\hat{F} - F_0) \xrightarrow{d} N(0, \omega^2)$	$n\hat{F} \xrightarrow{d}$ (noncentral) χ^2 mixture
noise/bias mean	$\text{tr}(\tilde{U}\Gamma)$, curvature-corrected	$\text{tr}(U\Gamma) = cdf$, no curvature
weight influence β_W	$O(\ e\)$, present	$\rightarrow 0$
ML vs GLS	differ (pseudo-true, β_W)	coincide to leading order
interval	Wald via ω^2 (16)	noncentral- χ^2 inversion

The local-asymptotics motivation is not wrong; it is a deliberate choice to work in the only regime where the df subtraction is exact and the curvature

and weight terms are negligible. Outside it, those terms are the leading description of a wrong model's fit.

6 ML, GLS, ULS

The three estimators share the quadratic-form weight at the data limit, $V = \frac{1}{2}D'(\Sigma_0^{-1} \otimes \Sigma_0^{-1})D$ for ML and GLS and $V = \frac{1}{2}D'D$ for ULS, but they differ in where the weight is anchored, and this is what separates their RMSEAs once the drift is gone.

ULS. Fixed weight, $\dot{V} = 0$, so $\beta_W \equiv 0$ and the bias is purely $\text{tr}(\tilde{U}T)/n$. ULS is never efficient, so $\text{tr}(UT) \neq df$ already at leading order regardless of normality; the curvature term is then a clean second correction with no weight bookkeeping. This is the cleanest place to exhibit $\tilde{U} \neq U$.

GLS. Weight $\hat{V} = V(s)$ at the sample moments, so $\dot{V}$ is the derivative of $V(\cdot)$ along the *raw* fluctuation z . The full β_W is present, leading-order in the residual through the cross term $2E[w'(I - PV)z]$, and ω^2 in (15) acquires the v_e term.

ML. The ML discrepancy is not a fixed quadratic; its estimating equation is the reweighted normal equation $\dot{\sigma}'V(\sigma(\theta))(\sigma_0 - \sigma(\theta)) = 0$ with the weight anchored at the *fitted model* $\Sigma(\theta)$. Two consequences. First, the pseudo-true value solves a model-metric projection, so under fixed misspecification $\theta_*^{\text{ML}} \neq \theta_*^{\text{GLS}}$ and $F_0^{\text{ML}} \neq F_0^{\text{GLS}}$: ML-RMSEA and GLS-RMSEA estimate *different population numbers*, already at zeroth order, because they measure misfit in different metrics (Olsson, Foss, Troye and Howell [2]). Second, the θ -dependence of the weight adds a discrepancy-curvature term K_F to B (a third derivative of F_{ML}), and routes the weight fluctuation through the projected increment $\dot{\sigma}(\hat{\theta} - \theta_*)$ rather than the raw z . All three effects collapse under correct specification ($\sigma(\theta_*) = \sigma_0$, $e = 0$), where ML, GLS and ULS coincide to the order that matters.

So the answer to “does GLS-RMSEA do anything that normal-theory RMSEA does not” is yes, and sharper than expected: under fixed misspecification the two already target different estimands (different metric), and on top of that carry different $O(1/n)$ biases through β_W and B . Under the drift, all of this disappears and the classical equivalence is restored. The daylight between estimators lives entirely in the fixed-misspecification, higher-order regime.

7 Numerical check

For ULS the weight is fixed, so $\beta_W \equiv 0$ and Corollary 1 predicts $N(\mathbb{E}[\hat{F}] - F_0) \rightarrow \text{tr}(\tilde{U}\Gamma)$ exactly, isolating the new object $\tilde{U}$. A one-factor model is fit to data from a two-factor population (six indicators, three per factor, factor correlation 0.55), a misspecification the one-factor model cannot absorb: $df = 9$, $F_0 = 0.231$, $\|e\| = 0.48$, curvature coupling $\|K\| = 0.68$. The table reports N times the Monte-Carlo bias of $\hat{F}$ against the naive $\text{tr}(U\Gamma)$ (Gauss–Newton A) and the curvature-corrected $\text{tr}(\tilde{U}\Gamma)$ (Hessian B). The sample covariance is the unbiased $\div(N-1)$ form; the $\div N$ form carries its own $O(1/N)$ moment bias which contaminates exactly this order. Under normality $\Gamma = \Gamma_{\text{NT}}$ is closed form; under t_6 data Γ is a Monte-Carlo estimate.

data	N	$N(\mathbb{E}[\hat{F}] - F_0)$ (MC se)	naive $\text{tr}(U\Gamma)$	corrected $\text{tr}(\tilde{U}\Gamma)$
normal	200	2.20 (0.05)	3.38	2.26
normal	800	2.24 (0.10)	3.38	2.26
t_6	200	4.26 (0.08)	7.00	4.74
t_6	800	4.60 (0.16)	7.00	4.74

Every cell matches the curvature-corrected trace and rejects the naive one: at $N = 800$ the empirical bias sits within one Monte-Carlo standard error of $\text{tr}(\tilde{U}\Gamma)$ in both data conditions, while $\text{tr}(U\Gamma)$ is wrong by the full gap (more than twenty standard errors at $N = 200$, normal). The small $N = 200$ shortfall in the t_6 cell is the expected $O(1/N)$ remainder, which closes by $N = 800$. The curvature term is not a third-decimal refinement: it is a third to a half of the entire bias. Script: `higher_order_discrepancy_check.py`.

8 Connections and caveats

Remark (Hessian bread, weight meat: the SE sibling). $\tilde{U}$ uses the population Hessian $B = A - K$ where U uses the Gauss–Newton A ; under correct specification $K = 0$ and they agree, the discrepancy-function analogue of the information equality. This is the same substitution that the misspecification-robust standard errors of `misspec_observed_bread.tex` make when they replace the Fisher bread by the observed Hessian, and β_W is the fit-index counterpart of their data-dependent-weight meat term. Both corrections are $O(\|e\|)$, hence zero at H_0 and leading-order off it: the order-promotion (Newey–McFadden null irrelevance, Hall–Inoue off-null promotion) is one phenomenon appearing in the variance there and in the bias here.

Remark (TIC analogy, one order deeper). `aic_tic_rmsea_bias.tex` reads df as the AIC penalty and notes the robust trace as its TIC analogue. Here $\text{tr}(UT)$ is that TIC-like trace (first order), and $\text{tr}(\tilde{U}T) + \beta_W$ is the next order. The user’s instinct stands: the df correction is AIC-like (a fixed integer), the robust trace is TIC-like (a sandwich trace), and the practical complaint about TIC, that $\text{tr}(\hat{I}^{-1}\hat{J})$ is noisy because $\hat{J}$ is a fourth-moment object, transfers verbatim: $\text{tr}(\hat{U}\hat{\Gamma})$ inherits the sampling variance of $\hat{\Gamma}$. The higher-order expansion is what lets one judge the trade: when $\|e\|$ and the kurtosis are small the AIC-like df is nearly unbiased and lower variance; when they are not, the bias of the cheap correction is exactly (13) and may dominate. A structured (model-implied or shrunk) $\hat{\Gamma}$ is the natural middle estimator, the RMSEA analogue of a regularized TIC.

Remark (Caveats). The truncation $\max(\cdot, 0)$ in (14) re-introduces an $O(n^{-1/2})$ bias near $F_0 = 0$, where the fixed-misspecification expansion is least relevant and the near-null device should take over. The ML term K_F and the projected weight fluctuation are stated, not derived; the full ML expansion needs the third derivative of F_{ML} and is left for the numerical companion. The expansion assumes finite fourth moments for Γ and finite sixth moments for the n^{-1} bias to have finite expectation; under heavier tails (13) is a stochastic-order statement, not a moment statement.

References

- [1] Browne, M. W. (1984). Asymptotically distribution-free methods for the analysis of covariance structures. *British Journal of Mathematical and Statistical Psychology*, 37(1), 62–83. doi:10.1111/j.2044-8317.1984.tb00789.x.
- [2] Olsson, U. H., Foss, T., Troye, S. V., & Howell, R. D. (2000). The performance of ML, GLS, and WLS estimation in structural equation modeling under conditions of misspecification and nonnormality. *Structural Equation Modeling*, 7(4), 557–595. doi:10.1207/S15328007SEM0704_3.
- [3] Satorra, A. (1989). Alternative test criteria in covariance structure analysis: A unified approach. *Psychometrika*, 54(1), 131–151. doi:10.1007/BF02294453.
- [4] Satorra, A., & Bentler, P. M. (1994). Corrections to test statistics and standard errors in covariance structure analysis. In A. von Eye & C. C. Clogg (Eds.), *Latent Variables Analysis* (pp. 399–419). Sage.

- [5] Shapiro, A. (1983). Asymptotic distribution theory in the analysis of covariance structures (a unified approach). *South African Statistical Journal*, 17(1), 33–81.
- [6] Shapiro, A. (2007). Statistical inference of moment structures. In S.-Y. Lee (Ed.), *Handbook of Latent Variable and Related Models* (pp. 229–260). Elsevier. doi:10.1016/B978-044452044-9/50013-9.
- [7] White, H. (1982). Maximum likelihood estimation of misspecified models. *Econometrica*, 50(1), 1–25. doi:10.2307/1912526.
- [8] Yuan, K.-H., Hayashi, K., & Bentler, P. M. (2007). Normal theory likelihood ratio statistic for mean and covariance structure analysis under alternative hypotheses. *Journal of Multivariate Analysis*, 98(6), 1262–1282. doi:10.1016/j.jmva.2006.08.005.